

Shivansh Karan

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EXPERIENCE

Agentic AI Automation Engineer | Emplay

Jan. 2026 – Jun. 2026

- Built document **ingestion and retrieval pipelines** for an enterprise procurement platform; structured catalog data was queried via **SQL** while unstructured KB content was searched using **vector similarity**, with results merged before the response layer.
- Implemented **NL-to-SQL** query generation with **expansion and retry logic**; when SQL returned empty results the pipeline **fell back to vector search** over the KB, reducing dead-end responses on structured enterprise queries.
- Developed workflow infrastructure for AI platforms, including **schema-aware retrieval pipelines**, **relationship-aware query orchestration**, ingestion callbacks, validation/retry mechanisms, and automation flows supporting enterprise scale data systems

Backend Engineer | Secnode

Jan. 2026 – Apr. 2026

- Built an **event-driven multi-agent platform** on **FastAPI & RabbitMQ**; a planner model decomposed user requests into subtasks, dispatched them to worker agents via message queues, and **streamed results over SSE** as each subtask completed.
- Implemented **structured event tracing** across the agent execution graph; each response was grounded in retrieved source documents with **citation tracking**, making agent outputs **fully auditable**, a hard requirement for security tool use cases.
- Built an **async report generation service** that aggregated agent task outputs into structured HTML/PDF reports; used a **dead-letter retry queue** so **partial agent failures didn't block report delivery** for the overall workflow.

AI Full Stack Developer | Anthrolic

Oct. 2025 – Nov. 2025

- Built an evaluation pipeline that ran **OCR** on uploaded handwritten answer images, scored each response by section against UPSC rubrics using a **structured LangChain prompt chain**, returning **per-criterion feedback** and a composite score.
- Designed the **async API** between the Next.js client and the scoring backend; LLM evaluation jobs ran in the **background** with a **polling endpoint** so the UI didn't time out waiting on slow inference calls.

Full Stack Developer | Anthrapi

Dec. 2024 – Sep. 2025

- Engineered **production-grade websites**, e-commerce platforms, landing pages, and **custom business management systems** for restaurants and SMB clients, delivering scalable full-stack solutions tailored to operational and branding requirements.
- Developed self-hosted **CRM/ERP-style platforms** with **workflow automation**, **admin dashboards**, and business process integrations, enabling streamlined operations, digital transformation, and improved client management workflows.

PROJECTS

Orchestrix | *FastAPI, PostgreSQL, Redis, Prometheus, Grafana, OpenTelemetry, Jaeger, Locust, Docker Compose, Lua*

[GitHub](#)

- Implemented a **multi-tenant job execution engine** handling concurrent job dispatch across distributed workers, enforcing per-tenant rate limits via a **Redis token bucket** with **atomic Lua scripts** to eliminate race conditions under high contention.
- Built a **durable job state machine** backed by Postgres with **optimistic locking** on status transitions and an async reaper process that recovers orphaned jobs from crashed workers, achieving **zero job loss** across **50K simulated executions**.
- Instrumented the full job lifecycle with **OpenTelemetry distributed tracing**, **Prometheus metrics**, and structlog, enabling end-to-end trace correlation from HTTP ingestion through worker execution with **sub-100ms P99 latency** on the hot path.
- Wrote a **Locust load test** simulating 10 tenants with different rate limits; caught and fixed a **head-of-line blocking bug** in the priority queue where high-traffic tenants were indefinitely delaying low-priority jobs under **burst load**.

CareerPilot | *LangChain, FastAPI, Gemini, pgvector, Playwright MCP, PostgreSQL, Supabase, Next.js*

[GitHub](#)

- Built an **asynchronous backend** for a career platform where resume uploads trigger a **multi-stage pipeline**: extracting structured fields, persisting to pgvector, performing **ANN job matching**, and queuing LLM-powered gap analysis and ranking.
- Designed the resume enrichment pipeline in discrete stages: deterministic parsing for field extraction, **pgvector vector similarity search** to retrieve similar job descriptions, then a final LLM pass that **flagged skill gaps** and ranked job matches.
- Integrated **Playwright MCP** with a **LangChain agent** to **automate job application form filling**; the agent parsed resume fields and mapped them to form inputs, handling layout and field-name variations across different job board sites.
- Designed the API layer to keep response times **under 200ms** by offloading resume enrichment to **FastAPI background workers**; clients polled a status endpoint and received results when the full pipeline completed.

TECHNICAL SKILLS

Languages: Python, TypeScript, Go

Backend & AI: FastAPI, Flask, Node.js, Asyncio, Pydantic, LangChain, LangGraph, gRPC

Databases & Messaging: PostgreSQL, Redis, MongoDB, RabbitMQ, Celery, Supabase, Pinecone, ChromaDB

Infra & DevOps: Docker, Nginx, OpenTelemetry, Prometheus, Grafana, Git, Pytest

EDUCATION

Dayananda Sagar College of Engineering, Bangalore

2022 – 2026

B.E. Computer Science and Engineering

ACHIEVEMENTS

ACM India Winter School 2025 – IIT Ropar: among the **40 students** selected nationwide.

Winner, Sentinel Hack 5.0: secured 1st place in the **Sentinel Hack 5.0 Hackathon**.

Runner Up, BMSCE Phaseshift: awarded 2nd place at Pragathi Idea Convention, **BMSCE Phaseshift**.